

Self-Distillation Without Imitation: Bootstrapping Reward Models with Privileged Information

Your Name

2026-06-29

Write your post here.

The idea is to use a privileged version of the model to evaluate and compare its own on-policy outputs, then distill those judgments into a reusable reward model for RL—learning from self-evaluation rather than imitating teacher responses.

Problems with Self-Distillation

We refer to (Tiwari et al. 2026) and (Kim et al. 2026), but briefly, here are some representative problems:

- **Privileged-information leakage:** the model learns to reference hints, feedback, or solutions unavailable at inference time.
- Diversity collapse
- **Suppression of epistemic reasoning:** see (Kim et al. 2026).

But why does ordinary OPD work?

Contrast: why a real teacher works

On-policy distillation from a stronger same-family model or an RL-trained expert succeeds because that teacher is close to the reward-tilted optimum: it puts more probability on genuinely high-reward responses while staying close enough for the student to copy, so the dense token-level signal points the same way the sparse RL reward would, just cheaper. The self-teacher fails precisely because privileged conditioning produces a teacher far from that optimum, pointing the dense signal at a response shape the student can't ground.

Method

SDRM-MSE

SDRM-BT

SDRM-GAIL

Apply RARO’s framework [^raro] (Cai and Provilkov 2025). The idea is to treat the teacher policy rollouts as expert demonstrations and then derive a reward signal from those expert demonstrations via IRL/GAIL.

[^raro] RARO (Relativistic Adversarial Reasoning Optimization) trains an LLM to reason from expert demonstrations without a hand-built verifier or preference labels by casting learning as an inverse-RL adversarial game. For each question, the policy generates an answer, while a reasoning critic compares it directly with the expert answer and predicts which is expert—or declares a tie. The critic receives reward for identifying the expert, and the policy receives reward when its answer is judged better or indistinguishable, so both are jointly updated with GRPO. In practice, RARO shares one LLM between the policy and critic, mixes both objectives in each training batch, and uses a replay buffer plus tie rewards to stabilize the continuously changing reward signal.

OPSD-Cartridge

Idea is to distill the PI into a cartridge.

Related work

Cai, Locke, and Ivan Provilkov. 2025. “Escaping the Verifier: Learning to Reason via Demonstrations.” *arXiv Preprint arXiv:2511.21667*.

Kim, Jeonghye, Xufang Luo, Minbeom Kim, Sangmook Lee, Dohyung Kim, Jiwon Jeon, Dongsheng Li, and Yuqing Yang. 2026. “Why Does Self-Distillation (Sometimes) Degrade the Reasoning Capability of LLMs?” *arXiv Preprint arXiv:2603.24472*.

Tiwari, Rishabh, Kusha Sareen, Devvrit Khatri, Lakshya Agrawal, Kurt Keutzer, Matei Zaharia, Inderjit S. Dhillon, and Rishabh Agarwal. 2026. “Why On-Policy Distillation Works and Naive Self-Distillation Doesn’t.” X Article. <https://x.com/rish2k1/status/2068414528598286485>.